

A proof of Zhao's conjecture on the log-concavity of $\{\sqrt[n]{T_n}\}_{n \geq 9}$ for the alternating sums of Motzkin numbers

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Abstract

Let $\{M_n\}_{n \geq 0}$ be the Motzkin numbers and $T_n = \sum_{k=0}^n (-1)^{n-k} M_k$ (OEIS A187306). We prove Conjecture 1 of F.-Z. Zhao [*The log-balancedness of the sequence for the alternating sums of Motzkin numbers*, INTEGERS **26** (2026), #A55]: the sequence $\{T_n^{1/n}\}_{n \geq 9}$ is log-concave. The proof establishes two-sided rational bounds for the ratios M_{n+1}/M_n and T_{n+1}/T_n by induction, reduces the conjecture to an explicit inequality for a logarithmic functional, and closes the argument with elementary classical inequalities together with exact computer verification of a finite list of polynomial and integer inequalities. All computational steps are certified in exact (rational or integer) arithmetic; no floating-point estimate enters the proof.

1 Introduction and statement

The Motzkin numbers $M_0 = M_1 = 1$, $M_2 = 2$, $M_3 = 4, \dots$ satisfy the classical recurrence

$$(n+3)M_{n+1} = (2n+3)M_n + 3nM_{n-1}, \quad n \geq 1. \quad (1)$$

Following Zhao [1], put

$$T_n = \sum_{k=0}^n (-1)^{n-k} M_k, \quad (2)$$

so $T_0, T_1, T_2, \dots = 1, 0, 2, 2, 7, 14, 37, 90, 233, 602, 1586, \dots$ (OEIS A187306). Zhao proved that $\{T_n\}_{n \geq 8}$ is log-balanced and posed the following conjecture.

Conjecture 1 (Zhao [1]). *For the sequence $\{T_n\}_{n \geq 0}$ defined by (2), $\{\sqrt[n]{T_n}\}_{n \geq 9}$ is log-concave.*

Writing $b_n = T_n^{1/n}$, the assertion is that $b_n^2 \geq b_{n-1}b_{n+1}$ for all $n \geq 10$. Our main result is:

Theorem 1. *Conjecture 1 holds. In fact $b_n^2 > b_{n-1}b_{n+1}$ for every $n \geq 10$.*

*This proof was produced in a two-agent collaboration set up by Jonathan Bostock. Adversarial review of the argument and of the verification scripts by OpenAI Codex is pending; until that review is complete this manuscript should be treated as a draft.

The strategy is the established one for n -th-root log-concavity of combinatorial sequences (cf. Luca–Stănică [2] and Chen–Guo–Wang [3] for the Motzkin and Catalan numbers themselves), executed here in a fully certified form: every analytic inequality is reduced either to a polynomial inequality on a real half-line, verified symbolically with an elementary certificate, or to an instance of $\log(1+x) \leq x$; every finite verification is done in exact integer or rational arithmetic. The verification scripts are `01_explore.py`, `02_design_bounds.py` and `03_sympy_proofs.py` in the directory `work/claude/zhao/` of the accompanying repository; the numbering (I1)–(I4), (C0)–(C7), and Parts A–E below matches the script output of `03_sympy_proofs.py`.

2 Preliminaries

From (2) we have immediately

$$T_{n+1} = M_{n+1} - T_n \quad \text{and} \quad T_n + T_{n-1} = M_n. \quad (3)$$

Let

$$x_n = \frac{M_{n+1}}{M_n}, \quad y_n = \frac{T_{n+1}}{T_n} \quad (\text{whenever } T_n \neq 0).$$

Dividing (1) by M_n , and (3) by T_n and using (3) once more,

$$x_n = \frac{2n+3}{n+3} + \frac{3n}{(n+3)x_{n-1}}, \quad (4)$$

$$y_n = \frac{M_{n+1}}{T_n} - 1 = x_n \left(1 + \frac{1}{y_{n-1}}\right) - 1, \quad (5)$$

where (5) is valid whenever $T_{n-1}, T_n > 0$; it is a rearrangement of equations (4)–(6) of [1]. Define

$$f_n(x) = \frac{2n+3}{n+3} + \frac{3n}{(n+3)x}, \quad g(x, y) = x \left(1 + \frac{1}{y}\right) - 1.$$

On $(0, \infty)$, f_n is strictly decreasing in x ; on $(0, \infty)^2$, g is strictly increasing in x and strictly decreasing in y . (Both statements are immediate.)

Lemma 2 (positivity; Part B). $T_n > 0$ for $n = 0$ and all $n \geq 2$.

Proof. Exact integer computation gives $T_n > 0$ for $2 \leq n \leq 220$ (Part B of the script). For $n \geq 220$ positivity follows from Theorem 3 below, whose induction (starting from the exactly verified $T_{40}, T_{41} > 0$) shows $y_n > 0$, hence $T_{n+1} = y_n T_n > 0$, for all $n \geq 40$. $\square$

3 Two-sided ratio bounds

For $n \geq 40$ define the rational functions

$$\begin{aligned} L_x(n) &= 3 - \frac{9}{2n} + \frac{10}{n^2}, & U_x(n) &= 3 - \frac{9}{2n} + \frac{15}{n^2}, \\ L_y(n) &= 3 - \frac{9}{2n} + \frac{7}{n^2}, & U_y(n) &= 3 - \frac{9}{2n} + \frac{22}{n^2}. \end{aligned} \quad (6)$$

(For orientation only: numerically $x_n = 3 - \frac{9}{2n} + \frac{c_x(n)}{n^2}$ with $c_x(n) \uparrow \frac{207}{16} = 12.9375$ and y_n likewise with $c_y(n) \uparrow \frac{225}{16} = 14.0625$, consistent with Kotěšovec's asymptotic $T_n \sim 3^{n+5/2}/(8\sqrt{\pi} n^{3/2})$; none of this is used in the proof.)

Theorem 3 (ratio bounds; Parts A and B). *For every $n \geq 40$,*

$$L_x(n) \leq x_n \leq U_x(n) \quad \text{and} \quad L_y(n) \leq y_n \leq U_y(n).$$

Proof. Induction on n . *Base case $n = 40$:* verified with exact rational arithmetic (Part B): $x_{40} = M_{41}/M_{40} \in [L_x(40), U_x(40)]$ and $y_{40} = T_{41}/T_{40} \in [L_y(40), U_y(40)]$; moreover $T_{40} > 0$, $T_{41} > 0$.

Induction step $n - 1 \rightarrow n$, $n \geq 41$: the four bound functions are positive on $[40, \infty)$ (certificates C0a, C0b for L_x, L_y ; $U_x \geq L_x$, $U_y \geq L_y$ termwise), so the intervals lie in $(0, \infty)$ and the monotonicity of f_n and g applies, and $T_n = y_{n-1}T_{n-1} > 0$ inductively. From (4) and $x_{n-1} \in [L_x(n-1), U_x(n-1)]$,

$$f_n(U_x(n-1)) \leq x_n \leq f_n(L_x(n-1)).$$

The script certifies (Part A), for all real $n \geq 41$,

$$U_x(n) - f_n(L_x(n-1)) \geq 0, \tag{I1}$$

$$f_n(U_x(n-1)) - L_x(n) \geq 0, \tag{I2}$$

whence $x_n \in [L_x(n), U_x(n)]$. Then from (5), $y_{n-1} \in [L_y(n-1), U_y(n-1)]$, and the bounds on x_n just obtained,

$$g(L_x(n), U_y(n-1)) \leq y_n \leq g(U_x(n), L_y(n-1)),$$

and the script certifies, for all real $n \geq 41$,

$$U_y(n) - g(U_x(n), L_y(n-1)) \geq 0, \tag{I3}$$

$$g(L_x(n), U_y(n-1)) - L_y(n) \geq 0, \tag{I4}$$

whence $y_n \in [L_y(n), U_y(n)]$. $\square$

Remark 4 (certificate methodology). Each of (I1)–(I4), (C0a)–(C7) is an inequality $\rho(n) \geq 0$ for a rational function ρ on a half-line $[n_0, \infty)$ ($n_0 = 41, 40$, or 100 as indicated). The script `03_sympy_proofs.py` verifies it with `sympy` in exact rational arithmetic as follows: write $\rho = P/Q$ over a common denominator; prove $Q > 0$ on $[n_0, \infty)$ (the leading coefficient is positive, `Poly.count_roots`(n_0, ∞) = 0, and $Q(n_0) > 0$); prove $P \geq 0$ on $[n_0, \infty)$ by substituting $n = n_0 + t$ and checking that *every* coefficient of the resulting polynomial in t is nonnegative. Every inequality listed in this paper passed through this all-nonnegative-shifted-coefficients path (see the script log); the weaker root-counting fallback coded in the script was never needed for a numerator. Since the certificates hold for all real $n \geq n_0$, they hold in particular for all integers $n \geq n_0$.

4 Reduction to a logarithmic functional

All logarithms are natural. For $n \geq 10$ the numbers T_{n-1}, T_n, T_{n+1} are positive by Lemma 2, so with $b_n = T_n^{1/n} > 0$,

$$b_n^2 \geq b_{n-1}b_{n+1} \iff \Phi(n) := n(n+1) \log y_{n-1} - n(n-1) \log y_n - 2 \log T_n \geq 0. \tag{7}$$

Indeed, $b_n^2 \geq b_{n-1}b_{n+1}$ is equivalent to $\frac{2}{n} \log T_n \geq \frac{1}{n-1} \log T_{n-1} + \frac{1}{n+1} \log T_{n+1}$; multiplying by $n(n^2-1) > 0$ gives $2(n^2-1) \log T_n - n(n+1) \log T_{n-1} - n(n-1) \log T_{n+1} \geq 0$, and substituting $\log T_{n-1} = \log T_n - \log y_{n-1}$, $\log T_{n+1} = \log T_n + \log y_n$ (note $n(n+1) + n(n-1) - 2(n^2-1) = 2$) yields (7). Strict inequality corresponds to strict inequality.

5 Exact verification for $10 \leq n \leq 110$

For $10 \leq n \leq 110$, raising $b_n^2 \geq b_{n-1}b_{n+1}$ to the power $2n(n^2 - 1) > 0$ shows it is equivalent to the integer inequality

$$T_n^{2(n^2-1)} \geq T_{n-1}^{n(n+1)} \cdot T_{n+1}^{n(n-1)}. \quad (8)$$

Proposition 5 (Part E). *Inequality (8) holds, strictly, for every integer $10 \leq n \leq 110$.*

Proof. Exact integer computation (Part E of the script). $\square$

6 The analytic bound for $n \geq 100$

For $k \geq 40$ set

$$u_k = \frac{3}{2k} - \frac{7}{3k^2}, \quad v_k = \frac{3}{2k} - \frac{22}{3k^2},$$

so that $L_y(k) = 3(1 - u_k)$ and $U_y(k) = 3(1 - v_k)$. The script certifies (C1–C6): $0 \leq v_k < 1$ for all real $k \geq 40$, and $0 \leq u_{n-1} < 1$ for all real $n \geq 100$. We use four consequences of the classical inequality $\log(1+x) \leq x$ ($x > -1$):

$$\log(1-t) \leq -t \quad (0 \leq t < 1), \quad (L1)$$

$$\log(1-t) \geq -\frac{t}{1-t} \quad (0 \leq t < 1), \quad (L2)$$

$$\frac{1}{k} \geq \log \frac{k+1}{k} \quad (k \geq 1), \quad (L3)$$

$$\sum_{k=m}^{n-1} \frac{1}{k^2} \leq \sum_{k=m}^{n-1} \frac{1}{k(k-1)} = \frac{1}{m-1} - \frac{1}{n-1} \leq \frac{1}{m-1} \quad (2 \leq m \leq n-1). \quad (L4)$$

((L2) is (L1)'s counterpart obtained from $\log(1+s) \leq s$ with $s = t/(1-t)$.)

Lemma 6 (ratio terms). *For all integers $n \geq 100$, with*

$$G(n) := -n(n+1) \frac{u_{n-1}}{1-u_{n-1}} + n(n-1)v_n,$$

we have

$$n(n+1) \log y_{n-1} - n(n-1) \log y_n \geq 2n \log 3 + G(n). \quad (9)$$

Proof. By Theorem 3, $y_{n-1} \geq 3(1 - u_{n-1})$ and $y_n \leq 3(1 - v_n)$. Since $\log$ is increasing, (L2) gives $\log y_{n-1} \geq \log 3 - \frac{u_{n-1}}{1-u_{n-1}}$ and (L1) gives $\log y_n \leq \log 3 - v_n$. Multiplying by $n(n+1) > 0$ and $-n(n-1) < 0$ respectively and adding (the $\log 3$ terms contribute $(n(n+1) - n(n-1)) \log 3 = 2n \log 3$) yields (9). $\square$

Lemma 7 (telescoped upper bound; C5, C6). *For all integers $n \geq 41$,*

$$2 \log T_n \leq 2 \log T_{40} + 2(n-40) \log 3 - 3 \log \frac{n}{40} + \frac{44}{117}. \quad (10)$$

Proof. $\log T_n = \log T_{40} + \sum_{k=40}^{n-1} \log y_k$, and by Theorem 3, (C5, C6) and (L1), $\log y_k \leq \log 3 + \log(1 - v_k) \leq \log 3 - v_k$ for $k \geq 40$. Now $\sum_{k=40}^{n-1} v_k = \frac{3}{2} \sum_{k=40}^{n-1} \frac{1}{k} - \frac{22}{3} \sum_{k=40}^{n-1} \frac{1}{k^2}$, and by (L3) the first sum is at least $\log(n/40)$ (telescoping $\sum \log \frac{k+1}{k}$), while by (L4) the second is at most $\frac{1}{39}$. Hence $-2 \sum_{k=40}^{n-1} v_k \leq -3 \log \frac{n}{40} + \frac{44}{3 \cdot 39} = -3 \log \frac{n}{40} + \frac{44}{117}$, which gives (10). $\square$

Lemma 8 (C7). $G(n) \geq -\frac{487}{40}$ for all real $n \geq 100$.

Proof. Certificate C7 (Remark 4); note $G(n) \rightarrow -\frac{27}{4} - \frac{22-7}{3} = -\frac{47}{4}$ as $n \rightarrow \infty$ and $\frac{487}{40} = 12.175 > \frac{47}{4}$. $\square$

Lemma 9 (final constant; Part D).

$$\Delta := 3 \log \frac{100}{40} + 80 \log 3 - 2 \log T_{40} - \frac{44}{117} - \frac{487}{40} > \frac{6}{5}. \quad (11)$$

Proof. Here $T_{40} = 49320944483427001$. Part D of the script bounds each logarithm by exact rationals: for rational $q > 1$ write $q = 2^k r$ with $r \in [1, 2)$, so $\log q = k \log 2 + \log r$, and for $s \in (1, 2]$ use $\log s = 2 \operatorname{artanh} \frac{s-1}{s+1} = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}$, $z = \frac{s-1}{s+1} \in (0, \frac{1}{3}]$: partial sums are exact rational *lower* bounds (all terms positive), and adding the geometric tail bound $\frac{z^{2K+3}}{(2K+3)(1-z^2)}$ gives exact rational *upper* bounds. Applying lower bounds to $3 \log 100 + 80 \log 3$ and upper bounds to $2 \log T_{40} + 3 \log 40$ yields the exact rational lower bound $\Delta \geq 1.2125 \dots > \frac{6}{5}$. $\square$

Proposition 10. $\Phi(n) > 0$ for every integer $n \geq 100$.

Proof. Combining (7), Lemma 6 and Lemma 7 (note $2n \log 3 - 2(n-40) \log 3 = 80 \log 3$),

$$\Phi(n) \geq G(n) + 3 \log \frac{n}{40} + 80 \log 3 - 2 \log T_{40} - \frac{44}{117}.$$

By Lemma 8 and since $\log(n/40)$ is increasing in n , for $n \geq 100$ the right-hand side is at least $\Delta > 0$ by Lemma 9. $\square$

7 Proof of Theorem 1

Proof. Let $n \geq 10$. If $n \leq 110$, then $b_n^2 > b_{n-1}b_{n+1}$ by Proposition 5. If $n \geq 100$, then $\Phi(n) > 0$ by Proposition 10, which by (7) is equivalent to $b_n^2 > b_{n-1}b_{n+1}$. The two ranges overlap and cover all $n \geq 10$, so $\{T_n^{1/n}\}_{n \geq 9}$ is (strictly) log-concave. $\square$

Remark 11 (computer verification summary). The verification script `03_sympy_proofs.py` (Python 3, `sympy`, exact `fractions.Fraction` and arbitrary-precision integers; no floating point in any certified step) performs: **Part A** = certificates (I1)–(I4) on $[41, \infty)$; **Part B** = exact base cases at $n = 40$ and $T_k > 0$ for $2 \leq k \leq 220$; **Part C** = certificates (C0a), (C0b) on $[40, \infty)$, (C1)–(C4) on $[100, \infty)$, (C5), (C6) on $[40, \infty)$, (C7) on $[100, \infty)$; **Part D** = exact rational log bounds for Lemma 9; **Part E** = the integer inequalities (8) for $10 \leq n \leq 110$. The script terminates with

OVERALL: ALL STEPS MACHINE-VERIFIED.

Supporting exploration (data to $n = 2099$, verification of Mathar’s four-term recurrence for A187306 — not used in the proof — and asymptotic fits) is in `01_explore.py`; the selection of the constants in (6) is documented in `02_design_bounds.py`.

References

- [1] F.-Z. Zhao, The log-balancedness of the sequence for the alternating sums of Motzkin numbers, *INTEGERS* **26** (2026), #A55.
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